# A simple implementation of Gadget for cod in Icelandic waters

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**Abstract:** Gadget is a statistical modelling framework which can be used to create multispecies, multifleet, multiarea models. The development of a complex multispecies model requires understanding of the modelled single species population. Simple single species models also are essential in evaluating whether more complex models increase understanding of the population dynamics. In this paper, Gadget is illustrated using a single species case study with two stock components on one area. Features of the implementation include immature and mature stock components, maturation, predation of the immature stock component and two fleets. Parameter estimation is done using maximum likelihood based on a variety of datasets. The estimated parameters relate to growth, maturation, fleet selection, recruitment, the initial population and consumption. Likelihood data include survey indices, biological samples from the catch and survey and stomach data. Optimisation of the model, along with a protocol to estimate appropriate weighting of the likelihood components are described. The effect of reduced data availability on parameter estimation is also considered, both by aggregating the likelihood component data (eg using coarser length categories) and by excluding age data from the likelihood components.

**Keywords:** Gadget, statistical model, maximum likelihood, predation, cod.

## 1 Introduction

Fishery science deals with the analysis of the interactions between fisheries and the ecosystem, possibly taking into account economic and social aspects. A computer program, Gadget, has been developed to undertake such analysis in as objective a manner as possible, using formal statistical modelling approaches. This paper gives a simple example of the use of Gadget to describe a species consisting of two stock components, immature and mature fish, in a single area. Extensions to more species and areas are described.

Data are routinely collected from the ecosystem, mainly by sampling and measuring fish in various types of surveys and from the fishery in the form of recording the catches, effort and biological sampling of the catches. These two types of biological samples routinely include measurements of the total amount caught, length measurements of individual fish, expensive age measurements from a subset of these and stomach content analysis from an even smaller subset.

Different models of the population dynamics of fish stocks have been able to utilise varying amounts of these data. In recent decades models have been developed to use all of the data through

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likelihood components when describing species and fleet interactions in spatially disaggregated models.

The Gadget environment has been developed to utilise as many of these data as possible, using appropriate assumptions on each data set.

# 2 Basic model setup

Models described in this paper are implemented using the computer program Gadget (Begley 2004), which is a derivative of Bormicon (Stefansson and Palsson 1997; Stefansson and Palsson 1998), which uses concepts developed in earlier work such as MULTSPEC (Tjelmeland and Bogstad 1989) and MSVPA (Helgason and Gislason 1979). Gadget can be used to implement a wide variety of ecosystem models with only a few possibilities occurring in the present case study.

The model is a parametric statistical model with parameters which can be fitted using maximum likelihood. Internally, the model tracks the number of fish by age and length within each area, time step and stock unit. The development of the ecosystem of all such units is deterministically projected or simulated forward in time resulting in a single realisation of the development of the ecosystem. A single such simulation results in a variety of information including stock trends, mean length and so on, for each stock unit. These simulation results can be compared to data through (negative logs of) likelihood functions. A search algorithm can subsequently be used to estimate values of the unknown parameters.

Denote by $N_{almsrt}$ the number ($N$) of fish of species $s$, age $a$, length $l$ and maturity stage $m$, alive in region $r$ and timestep $t$. This group of fish is a subset of the entire collection of fish of a given species. Such a collection will be referred to as a subpopulation (or substock). Mathematically, this is just a collection of numbers, but they are usually unknown and will need to be estimated using statistical techniques.

Normally there are only two maturity stages within the model, immature and mature fish ($m = 0, 1$). In many cases only a fixed species, area, time step and maturity stage are considered and in this case such indices are omitted and the notation is simplified to e.g. $N_{al}$.

In order to distinguish between predator and prey species in an interaction, the following notation is used:

| | |
|---|---|
| $s$ | a general species. |
| $p$ | a prey species. |
| $P$ | a predator species. |
| $l$ | a prey length group. |
| $L$ | a predator length group. |

Accordingly, $N_{lp}$ denotes the number of individuals of a prey species $p$ of length $l$ and $N_{LP}$ the numbers in length group $L$ of a predator $P$. The notation is solely for convenience as there is no reason mathematically to exclude one predator's prey from being another species' predator.

Several processes can affect such a group of fish and cause a change in numbers in the group. The change in numbers can be either due to a transfer of fish between groups or mortality. The various processes available as model components in Gadget are listed here, but a detailed description of components used in the present case study is given in the following section.

- Migration:

  Migration is not used in the present case study. The migration a subpopulation undertakes on a given time step can be described by matrices, $A_{almst} = (a_{almsr_1r_2t})_{r_2=1,\ldots,R}^{r_1=1,..R}$, containing

the proportion $a_{almsr_1r_2t}$ of the population of the subpopulation that moves from area $r_2$ to area $r_1$ and $R$ is the number of areas. Hence, if $u = (u_1, ..., u_R)$ are abundance numbers by area for an age-length-maturity-species group in a subpopulation, the area distribution after the migration has taken place, is $A_{almst}u$.

- Maturation:

  Maturation involves shifting fish from a subpopulation of immature fish to a subpopulation of mature fish. This is done using proportions (as in the migration process), which are designed to mimic the resulting proportion of mature fish in each age-length cell.

- Consumption:

  Predation causes mortality in the prey stock and can be used to determine the growth of the predator.

- Individual growth:

  Within population dynamics models growth in length may be implemented through growth in weight since the weight increase is more naturally linked to consumption. In this case, the target weight increase, $\Delta W$ is a function of consumption and fixing a length-weight relationship of the form $W = aL^b$ can give an approximate average length increase through e.g. a Taylor approximation. Alternatively, as used in the present case study, the length growth is according to a specified functional form and a length-weight relationship is used to implement growth in weight.

  Regardless of which method is used to compute $\Delta W$, the length increase $\Delta L$ needs to be implemented through an update mechanism which moves fish between length groups in such a fashion that the average movement corresponds to a length growth of $\Delta L$. Growth in length is therefore modelled through the use of growth update matrices. The matrices are described in a parametric manner but the resulting growth needs to match the average growth of $\Delta L$.

- Natural mortality:

  In addition to predation/fishing, "other" natural mortality is implemented in the usual fashion, viz

  $$N_{a,t+1} = e^{-M_a \Delta t} N_{a,t}.$$

- Ageing:

  The last time step of a year involves increasing the age by one year, except for the oldest age group which is a plus group.

- Spawning:

  A (mature) subpopulation may generate offspring and lose biomass. This possibly results in spawning mortality.

# 3 A case study

The primary case study considered in the present paper considers two stock components of cod, i.e. mature and immature cod in Icelandic waters which are considered a single area. This will illustrate biological assumptions which need to be made and several statistical estimation issues.

Since mature cod consume immature cod, the present setup will illustrate statistical incorporation of stomach content data into the estimation procedure, as well as a variety of other data sources.

The emphasis of the case study is the illustration of complex estimation issues and how these can be solved. It is important to note that using simpler models tends to ignore these issues rather than solve them. This becomes particularly obvious when Gadget is used for estimation, since the approach permits the user to either estimate or fix values of parameters and as it turns out, values for, or relationships between parameters do need to be assumed since they are in some cases very poorly determined.

Several important multispecies interactions are known to be of importance in the marine ecosystem off Iceland (Stefansson et al. 1998b).

This particular case study does not take into account capelin, the major food source of cod. Hence the effect of consumption on the growth of the predator will not be included as a part of the model, though this is known to be an important factor (Magnusson and Palsson 1991; Steinarsson and Stefansson 1991). For the same reason, temperature or other environmental factors are not included (though they certainly could be). The environmental effects are known to exist and be considerable (Stefansson et al. 1998b) and should be included, at least as a factor affecting metabolism if capelin is included, though these effects may also be taken into account to describe catchability changes (Stefansson and Eiriksson 1998).

On the other hand, predation as a source of mortality can be important (Bogstad et al. 1994) and this is taken into account in the present model, but only with regard to the predation of mature cod on immature cod. Other factors, such as marine mammals may potentially be important (Stefansson et al. 1998a).

# 4 The biological model: Component processes

## 4.1 Growth

For immature cod, growth is assumed to be according to a von Bertalanffy equation (1).

$$L_t = L_\infty \left(1 - e^{-Kt}\right), \; t \in \mathbf{R}_+, \tag{1}$$

Suppose fish of a given age, $a$, happen to be of length $l$. The growth is given by

$$\begin{aligned} \Delta L &= L_\infty \left(1 - \frac{L}{L_\infty}\right)\left(1 - e^{-K\Delta t}\right) && (2)\\ &\simeq K\left(L_\infty - L\right)\Delta t, && (3) \end{aligned}$$

where the approximation is either based on the Taylor approximation (3) to the exponential function or, alternatively, the approximation is simply based on discretising an appropriate differential equation. Although the approximation is commonly used, it is not quite adequate for general use in Gadget since it is virtually impossible to simulate exact processes using the approximation and hence the exact formula (2) will be preferred to (3) in this case study.

In the present paper the growth in weight and growth in length will be assumed to be linked and a length-weight relationship of the form

$$W = cL^b \tag{4}$$

is used as a basis. In this relationship, $c$ becomes the condition factor, $c = W/L^3$ when $b = 3$ but $c$ has little direct meaning otherwise.

It should be noted that when $l$ is greater than $L_\infty$, growth should become negative according to the growth equation. The implementation of the equation in Gadget does not allow for $l$ to be greater than $L_\infty$ which should exceed the maximum length defined in the model.

Also, the average growth of a group of fish next needs to be translated into a new distribution of these fish in a subsequent time step. This will be handled in subsequent sections.

### 4.1.1 Updating length distributions

Fish population dynamics are modelled in MULTSPEC and Gadget through forward simulations of fish populations, allowing fish to migrate between areas, die, grow, mature and spawn. The basic unit in these models is the number of fish in a certain model unit or "cell". The fish in a "cell" are in the same age and size group, in the same region and time step. When this basic model formulation is used, the numbers in a "cell" need to be updated during a given time step, so as to reflect all processes being modelled. In addition to growth, these processes include migration, spawning, natural and fishing mortality.

Starting with a specified average length increment ($\Delta L$) fish need to be distributed from their initial length class into upper length classes in a reasonable manner. Simple techniques may use only few upper length intervals and use a simple ad-hoc update scheme. The update scheme should eventually be evaluated in terms of its ability to provide adequate eventual length distributions. This sets some immediate bounds on the dispersion at each time step, since an overly high or low variance in the length update will quickly result in inadequate final length distributions at age for the oldest ages.

The length update scheme can most easily be implemented through a look-up table, where a discrete set of $\Delta L$-values is provided along with the distribution to be used for reallocating the length group when the chosen growth is $\Delta L$. This approach, used in earlier Bormicon, MULTSPEC and Fleksibest implementations (e.g. Stefansson and Palsson (1997)), is undesirable for many reasons. Firstly, the setup is completely rigid as there is no built-in parameter to describe possible deviations of growth from the specified distribution and hence data on growth may adversely affect parameters in other parts of a complex model only because of incorrect specification of the rigid relationship. Secondly, a simple discrete (rounded) lookup provides a nondifferentiable likelihood function which will result in estimation problems later on.

What is needed is a way to specify a flexible parametric distribution with enough parameters to allow minimal flexibility to track length distributions of an age group, yet with enough parsimony in parameters to allow for the estimability of the parameters involved.

Consider fish of a specified length, $L$, which are destined to grow on average $\Delta L$, according to the growth model. A model for the update will assign probabilities, $p_j$ of a fish in the original cell growing by $j$ length groups. Denote the absolute length increment corresponding to $j$ length groups by $\delta l_j$. These probabilities must then satisfy the obvious restrictions

$$\begin{aligned} \sum_j p_j &= 1 \\ \sum_j p_j \delta l_j &= \Delta L. \end{aligned}$$

Since the present approach is length-based, the weight needs to be updated in synchrony with the length increase. Consider therefore fish of length $L$ with an average weight of $W$, which are due to increase in length by $\delta l_j$ from length $L$. Although it is possible to use Taylor approximations to the length-weight relationship, as implemented in Bormicon, a more internally and mathematically

consistent approach is to maintain length-weight relationships exactly. This can be done by defining the upcoming weight increment of this group of fish as

$$\delta w_j \quad = \quad c\,(L + \delta l_j)^b - cL^b.$$

The mean weight in the receiving group is subsequently updated using prior and incoming numbers and mean weight in the group.

The result of this approach is that if fish start out so that the length-weight relationship holds exactly for all length cells, then this relationship is maintained throughout the simulation. In the present case study, however, the initial weight at length is determined from smoothed data and hence the length-weight relationship is not precisely maintained.

## 4.2 A formal model for the update

Although a first step might be to attempt to estimate individual probabilities, $p_j$, these would result in far too many parameters. Another approach would be to estimate variance, skewness and kurtosis and go from these to transition probabilities, but there is no trivial transformation between the two.

A flexible distribution such as the 4-parameter inverse lambda distribution could probably be used (Ramberg et al. 1979), but parameter estimation tends to be difficult. Similarly, a binomial distribution (or even a (truncated) Poisson) can be used, but both are completely rigid, since the value of $\Delta L$ completely specifies the single free parameter in each of these distributions (assuming the number of permissible length group increments to be fixed).

The beta-binomial distribution can be used as a simple alternative. This approach can be formulated so as to provide a single estimable parameter in addition to the mean, which is specified by $\Delta L$.

First consider the binomial distribution which is defined for integers, $x = 0, \ldots, n$ by

$$\binom{n}{x} p^x (1-p)^{n-x} = \frac{\Gamma(n+1)}{\Gamma(x+1)\Gamma(n-x+1)} p^x (1-p)^{n-x}$$

Using this distribution for the issue at hand, for a given $n$, the other parameter, $p$, of this distribution, is fully defined since $\mu = np$ and the mean growth is given as the specified $\Delta L$, which fixes $p = \Delta L/n$. Although this distribution can certainly be used, it is clear that no flexibility is allowed at all and in fact it would be quite unlikely for such a rigid distribution to satisfy the specified requirements of attaining the correct final distribution of length at age.

A common approach to more flexibility is to allow the parameter $p$ itself to come from another distribution, often the beta distribution.

The beta distribution is defined for arbitrary values of $\alpha > 0$ and $\beta > 0$ by

$$f(p) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1}(1-p)^{\beta-1}, \quad 0 \le p \le 1 \tag{5}$$

and it is well-known that the mean of this distribution is given by

$$E(p) = \frac{\alpha}{\alpha+\beta} \tag{6}$$

Thus, rather than using a fully specified binomial distribution, more flexibility is obtained by using this combined beta-binomial distribution. This approach results in the following marginal distribution of the length increments:

$$\begin{aligned} P[X=x] &= \int_{p=0}^{1} P[X=x|p]f(p)dp \\ &= \int_{p=0}^{1} \frac{n!}{x!(n-x)!}p^x(1-p)^{n-x}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}p^{\alpha-1}(1-p)^{\beta-1}dp \\ &= \frac{\Gamma(n+1)\Gamma(\alpha+\beta)}{\Gamma(x+1)\Gamma(n-x+1)\Gamma(\alpha)\Gamma(\beta)}\int_{t=0}^{1} p^{x+\alpha-1}(1-p)^{n-x+\beta-1}dp \\ &= \frac{\Gamma(n+1)\Gamma(\alpha+\beta)}{\Gamma(x+1)\Gamma(n-x+1)\Gamma(\alpha)\Gamma(\beta)}\frac{\Gamma(x+\alpha)\Gamma(n-x+\beta)}{\Gamma(n+\alpha+\beta)} \\ &= \frac{\Gamma(n+1)}{\Gamma(n-x+1)\Gamma(x+1)}\frac{\Gamma(\alpha+\beta)}{\Gamma(n+\alpha+\beta)}\frac{\Gamma(n-x+\beta)}{\Gamma(\beta)}\frac{\Gamma(x+\alpha)}{\Gamma(\alpha)} \end{aligned}$$

For any positive number, $y$, the relationship $\Gamma(y+1) = y\Gamma(y)$ holds. It follows that for $\alpha, \beta > 0$, any integer $n \geq 1$ and $x = 0, 1, \ldots, n$ the following relationships hold

$$\frac{\Gamma(x+\alpha)}{\Gamma(\alpha)} = \begin{cases} 1 \text{ for } x = 0 \\ \alpha \text{ for } x = 1 \\ (x-1+\alpha)(x-2+\alpha)(x-3+\alpha)\cdot\ldots\cdot(\alpha+1)\alpha \text{ for } x > 1 \end{cases}$$

and

$$\frac{\Gamma(n-x+\beta)}{\Gamma(\beta)} = \begin{cases} 1 \text{ for } n-x = 0 \\ \beta \text{ for } n-x = 1 \\ (n-x-1+\beta)(n-x-2+\beta)(n-x-3+\beta)\cdot\ldots\cdot(\beta+1)\beta \text{ for } n-x > 1 \end{cases}$$

and finally, for $n \geq 2$:

$$\frac{\Gamma(\alpha+\beta)}{\Gamma(n+\alpha+\beta)} = \frac{1}{(n-1+\alpha+\beta)(n-2+\alpha+\beta)(n-3+\alpha+\beta)\cdot\ldots\cdot(1+\alpha+\beta)(\alpha+\beta)}$$

It follows that for $n \geq 1$ the above probabilities can be rewritten as

$$P[X=x] = \begin{cases} \frac{(n-1+\beta)(n-2+\beta)\cdot\ldots\cdot(\beta+1)\beta}{(n-1+\alpha+\beta)(n-2+\alpha+\beta)(n-3+\alpha+\beta)\cdot\ldots\cdot(1+\alpha+\beta)(\alpha+\beta)} & \text{for } x = 0 \\ n\cdot\frac{(n-2+\beta)(n-3+\beta)\cdot\ldots\cdot(\beta+1)\beta}{(n-1+\alpha+\beta)(n-2+\alpha+\beta)\cdot\ldots\cdot(1+\alpha+\beta)(\alpha+\beta)}\cdot\alpha & \text{for } x = 1 \\ \frac{\{n\cdot(n-1)\ldots\cdot(n-x+1)\}}{x!} & \\ \quad\cdot\ \frac{(n-x-1+\beta)(n-x-2+\beta)(n-x-\ 3+\beta)\cdot\ldots\cdot(\beta+1)\beta}{(n-1+\alpha+\beta)(n-2+\alpha+\beta)(n-3+\alpha+\beta)\cdot\ldots\cdot(1+\alpha+\beta)(\alpha+\beta)} & \\ \quad\cdot\ (x-1+\alpha)(x-2+\alpha)(x-\ 3+\alpha)\cdot\ldots\cdot(\alpha+1)\alpha & \text{for } 2 \leq x \leq n-2 \\ n\cdot\frac{(n-2+\alpha)(n-3+\alpha)\cdot\ldots\cdot(\alpha+1)\alpha}{(n-1+\alpha+\beta)(n-2+\alpha+\beta)\cdot\ldots\cdot(1+\alpha+\beta)(\alpha+\beta)}\cdot\beta & \text{for } x = n-1 \\ \frac{(n-1+\alpha)(n-2+\alpha)\cdot\ldots\cdot(\alpha+1)\alpha}{(n-1+\alpha+\beta)(n-2+\alpha+\beta)\cdot\ldots\cdot(1+\alpha+\beta)(\alpha+\beta)} & \text{for } x = n \end{cases}$$

It should be noted that in the case of growth by length groups within a short time interval, only low values of $n$ and $x$ are needed and the above products become correspondingly simple to generate.

Thus, probabilities can readily be generated from this beta-binomial distribution, given specified values of $\alpha$, $\beta$ and $n$. The last parameter of these, $n$, will usually be assumed, outside an estimation procedure.

It is also reasonably easy to see that the mean of the beta-binomial distribution is given by $\mu = nE[p] = \frac{n\alpha}{\alpha+\beta}$. If $\beta$ is taken as a parameter to be estimated, the requirement $\Delta L = \mu$ therefore implies $\alpha = \frac{\beta \Delta L}{n - \Delta L}$.

This approach will therefore be implemented in Gadget by defining a new growth function with a single estimable parameter $\beta$, to be set (along with $n$) to an initial value in a specification file.

### 4.3 Alternative growth formulations

An alternative set of growth functions is obtained by basing length growth on weight growth and computing weight growth from consumption. In order to verify different methods of implementing growth, it is useful to be able to link these approaches together and this can be done in Gadget (Begley 2004).

The general case will involve consumption and therefore growth in length and growth in weight will not always conform strictly to a length-weight relationship. Notably, if fish do not get enough food, they will typically lose weight but not get shorter. This effect can be accounted for within Gadget but is not implemented in the present case study.

## 5 Predation, including fishing and consumption

The consumption of a prey by a predator is defined through the following set of equations. Consider a fixed area and time step so the notation can be simplified to indicate only the length of predator and prey.

The suitability function, $S_{P,p}$ is viewed as a function of two variables, the length of the predator and prey.

$S_{P,p}(l, L)$ the suitability of length group $l$ of prey $p$ as food for length group $L$ of predator $P$.

As a function of $l$, over the domain of prey length groups, the form of this function can be for example, strictly increasing or dome shaped. The function needs to reflect the relative desirability of one prey size group to another for a given predator size group.

In this case study the suitability function is given by

$$S_{P,p}(l, L) = \begin{cases} p_0 + p_2 e^{-\frac{(\ln \frac{L}{l} - p_1)^2}{p_4}} & \text{if } \ln \frac{L}{l} \leq p_1 \\ p_0 + p_2 e^{-\frac{(\ln \frac{L}{l} - p_1)^2}{p_3}} & \text{if } \ln \frac{L}{l} > p_1, \end{cases} \tag{7}$$

with five parameters allowing an asymmetric relationship about the optimal predator prey size ratio.

Define the auxiliary function $\phi_P$ by

$$\phi_P(l, L, p) := S_{P,p}(l, L) N_p(l) W_p(l), \tag{8}$$

which simply weights the prey biomass according to its suitability. This is used to define the *feeding level*

$$\psi_P(L) := \frac{\sum_{l,p} \phi_P(l, L, p)}{\sum_{l,p} \phi_P(l, L, p) + A}, \tag{9}$$

where $A = A_1 A_2$ is a constant, commonly separated into the size of the area, $A_1$ and a constant $A_2$ representing the prey density at which consumption is half the maximum consumption.

The *target consumption*, i.e. the amount which length group $L$ of $P$ will consume of length group $l$ of $p$, assuming that enough food is available, is then given by the equation

$$\tau_{P,p,L,l} = N_P(L) H_P(L, T) \psi_P(L) \frac{\phi_P(l, L, p)}{\sum_{l,p} \phi_P(l, L, p)}, \tag{10}$$

where $T$ denotes area-specific temperature and the function $H$ is termed the *maximum consumption* which is given by

$$H_P(L, T) = m_0 e^{(m_1 T - m_2 T^3)} L^{m_3}. \tag{11}$$

For each prey an upper limit needs to be set on the total amount consumed by all predators so as not to obtain more consumption than available biomass. Consumption is limited to 95% of the available biomass. This is implemented by scaling target consumption by all predators through the understocking likelihood component, given by the equation

$$\ell = \sum_{time} \sum_{areas} \Big( \sum_{predators} U \Big)^g,$$

where $U$ is the understocking that has occurred in the model and $g$ is a constant.

Commercial catches can be implemented using the predation procedure, by setting the *target catch* directly equal to the observed catch and using the same scaling mechanism as for (other) predators when prey biomass is too low to sustain total predation (including catches).

Alternatively, commercial or survey catches can be set to be linearly related to the abundance numbers of the preys. In this case, the amount caught of length group $l$ of the prey $p$ on the area $a$ and time step $t$ by fleet $P$ is

$$\begin{aligned} \tau_{P,p,L,l} &= c E_{t,a} \phi_P(l, L, p) && (12) \\ &= c E_{t,a} S_{p,P}(l, L) N_p(l) W_p(l) && (13) \end{aligned}$$

where the effort measure $E_{t,a}$ is a parameter, c is a constant and $L$ is a number appropriate for scaling the suitability function into a selection pattern for the fleet.

# 6 Available raw and basic data summaries

## 6.1 Data collection

Biological data are collected by the Marine Research Institute as a part of a standard monitoring programme and data on landings are collected by the Directorate of Fisheries. In both cases all data are stored in Oracle data bases. The data of interest to the current case study are:

- Landings data by species, fleet and month.
- Age, length and maturity measurements from surveys and biological sampling of the catch.
- Stomach data
- Survey indices

## 6.2 Data Warehouse

Data input to Gadget is normally through a data warehouse, which is used to store data at a minimally aggregated level (i.e. highly disaggregated) and described in Chapter Kupca (2003). In this case study data from the MRI data base are aggregated onto the subdivisions set out in Section Taylor (2003). Aggregations are through simple arithmetic summaries (averages or sums, as appropriate). Input data files and likelihood data files can be extracted from the data warehouse with the user defining the level of aggregation to be output.

# 7 Data and likelihood functions

As indicated in Stefansson (1998), a statistical multispecies model can be based on a variety of different data summaries, but care must be taken not to use the same information twice, e.g. both through mean length at age and as length distributions with age-length keys.

It is important to aggregate the data to a level at which coverage of the selected length/age interval is adequate as zeros are taken to mean zero. Timesteps with low sampling or sampling restricted to a limited range of length/age groups may result in zeros which are an artifact of sampling and unrepresentative of the population. This is true for all likelihood components.

The data used in the current analysis consist of measurements from commercial catches and the two main groundfish surveys. Not all months have measurements for all fleets and in particular the groundfish surveys each occur only in a single month. Each of these data sets is linked to the model through a likelihood component. Given the problems inherent in mixing likelihood components corresponding to different probability distributions and in particular the known problems with a multinomial distribution being inadequate for length distributions, only normal distributions are used and each negative log likelihood component is simply a (weighted) sum of squares.

Data used in the case study are:

Input data:

- Landings data are obtained from the Directorate of Fisheries data base.

Biological sampling likelihood data:

- Length distribution, aggregated on 2 cm intervals, from sea and harbour sampling of commercial catches by month from 1984 to 2002.
- Length distribution, aggregated on 2 cm intervals, from the March groundfish survey (Palsson et al. 1989) from 1985 to 2002.
- Age length composition, aggregated on 4cm intervals, from commercial catches by month from 1984 to 2002.
- Age length composition, aggregated on 4 cm intervals, from the March groundfish survey (Palsson et al. 1989) from 1985 to 2002.
- Age composition from the March groundfish survey (Palsson et al. 1989).
- Age composition of the commercial catches by month from 1984 to 2002.

These likelihood data are of the same class – biological sampling – with all ages aggregated for the length distributions and all lengths aggregated for the age distributions. The sum of squares likelihood function is

$$\sum_{time} \sum_{areas} \sum_{ages} \sum_{lengths} (p_{tral} - \pi_{tral})^2$$

where
$p$ is the proportion of the data sample for that time/area/age/length combination
$\pi$ is the proportion of the model sample for that time/area/age/length combination.

Stock composition likelihood data:

- Number immature and number mature at length in 2cm length classes from the March groundfish survey (Palsson et al. 1989).

This groundfish survey is undertaken at a time when the maturity stage is easily determined and is unaffected by the potential biases due to targeting of the mature stock component by the commercial fleet. Although the maturation process in Gadget can be age and length dependent, in this case study only length is taken into account and ages are aggregated.

The sum of squares likelihood function for this component is

$$\sum_{time} \sum_{areas} \sum_{ages} \sum_{lengths} \sum_{stocks} (p_{tral} - \pi_{tral})^2$$

where
$p$ is the proportion of the data sample for that time/area/age/length/stock combination
$\pi$ is the proportion of the model sample for that time/area/age/length/stock combination.

Survey index likelihood data:

- Three survey indices from the March survey (Palsson et al. 1989) (figure 1) with one datum for each group for the years 1985–2002.
- Three survey indices from the autumn survey (Sigurdsson et al. 1997) (figure 1) with one datum for each group for the years 1995–2002.

Three indices are calculated for each survey by splitting the length distribution into three groups with the division based on the 'typical' structure of the length distribution for the survey over all years. These groups represent age 1, age 2 and age 3 and older.

The likelihood component for these data is the sum of squares of a log linear regression with the slope and intercept estimated or with the slope fixed and only the intercept estimated.

$$\ell = \sum_{time} \Big( \ln(I_t) - \big(\alpha + \beta \ln(N_t)\big) \Big)^2$$

where $I$ is the observed survey index and $N$ is the corresponding index calculated in the model.

In this case study, the slope and intercept are estimated for length groups 1 and 2 with the slope fixed to 1 for the third length group.

In figures 1 & 2 it can be seen that for length groups 1 and 2, the surveys provide similar information for the overlapping years (correlations of 0.59 and 0.92 for length groups 1 and 2 respectively) but there is conflicting information from the third length group (correlation of -0.15), which could be due to differences in the spatial distribution or depth in the water column of the mature stock component at the time of the surveys.

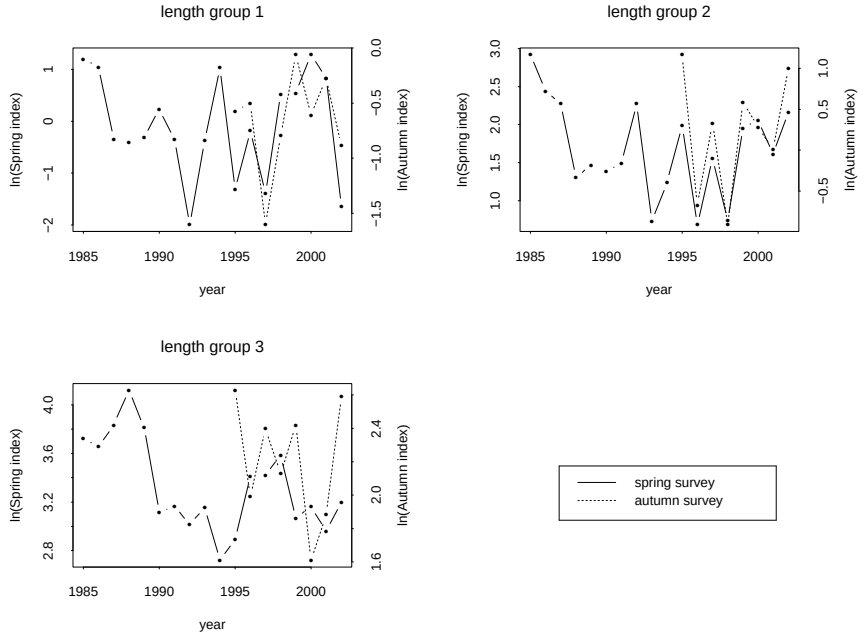



Figure 1: Survey indices for the spring and autumn surveys for each length group, with the y axis for the spring survey on the left and for the autumn survey on the right.

For both the spring and autumn surveys there is a positive relationship between length group 1 and 2 indices, as indicated in figure 7) with correlations of 0.84 and 0.66 for the spring and autumn surveys respectively.

Stomach content likelihood data:

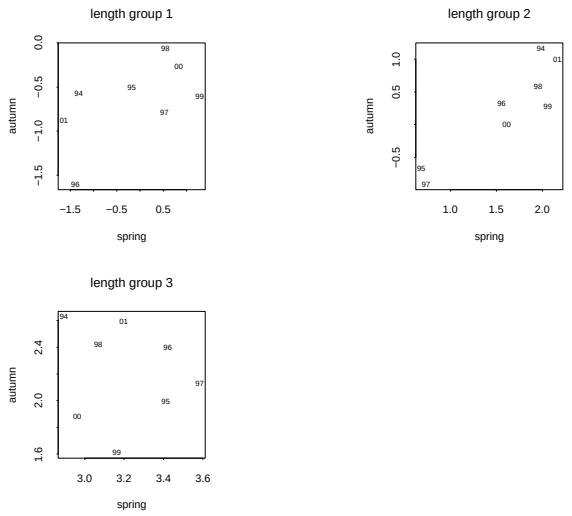



Figure 2: Scatter plots of the autumn survey vs the spring survey for each length group on a log scale.

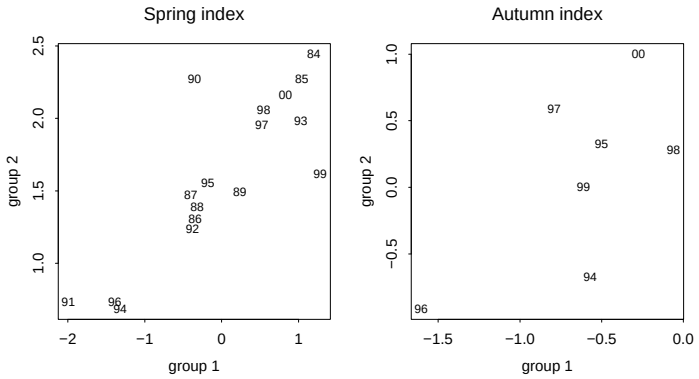



Figure 3: Scatter plots of the length group 2 index against the length group 1 index for each survey on a log scale with each point labelled with the cohort year.

- Composition of stomach contents as ratios of prey biomass by species (i.e. cod or other) aggregated over all predator lengths from the March groundfish survey for 1985–2002.
- Composition of stomach contents ratios of prey biomass by species (i.e. cod or other) aggregated over all predator lengths from other stomach sampling projects.
- Length distribution of prey by predator length groups as biomass ratios from the March groundfish survey for 1985–2002. Predators are aggregated into 10cm length groups and

prey into 4cm length groups.

- Length distribution of prey by predator length groups as biomass ratios from other stomach sampling projects. Predators are aggregated into 10cm length groups and prey into 4cm length groups.

The likelihood component for the stomach data biomass ratios is given by

$$\sum_{time}\sum_{areas}\sum_{prey}\sum_{L}\sum_{l}(p_{prey,L,l}-\pi_{prey,L,l})^2$$

where
$p$ is the proportion of the data sample for that time/area/prey/predator size group/prey size group combination
$\pi$ is the proportion of the model sample for that time/area/prey/predator size group/prey size group combination

Species biomass ratios and prey length distributions are available for all years of the spring groundfish survey. Data other than those from the spring survey are available for 39 timesteps from 1994 onwards with prey length distributions for all but 2 of the steps with cod prey recorded.

It should be noted that a particularly simple sum of squares is used as the log likelihood in all cases. There are several reasons for this, all somewhat technical (e.g. Elvarsson 2004).

# 8 Model Implementation

**?**)

The model runs from 1984 to 2002 on one area with two cod components – immature and mature. The immature are aged 1 to 10 and the mature ages 3 to 12, with the final age a plus group. Immature fish mature according to the maturation function and any not mature by the end of their tenth year move into the mature component. A single commercial fleet operates.

- Length growth is defined by equation 2 with $k$ estimated within Gadget and $L_\infty$ fixed to a value estimated directly from sampled catch and survey data. The betabinomial parameter $\beta$ is fixed to 1000 with maximum length group growth $n = 10$.

  Weight growth is according to equation 4 with different weight length relationships for the immature and mature components. The values of $c$ and $b$ were calculated from survey data and subsequently fixed.

- The function for maturation is:

$$M(l_t, a_t) = \frac{1}{1 + e^{-\alpha-\beta l_t-\gamma a_t}} \tag{14}$$

  As maturation in this case study is only a function of length, the age parameter $\gamma$ is set to zero. $\beta$ the length of 50% mature and the rate $\alpha$ are estimated within the model.

- Mortality at age is fixed.

- The mean and standard deviation of length of the initial populations were calculated from survey data and output from the datawarehouse. The weight-length relationships used for weight growth were also used for the initial population.

- The number at age of the initial population (ages 2 to 12) is estimated within the model, with the age distribution estimated and then a multiplier used to scale the population abundance.
- Mean and standard deviation of length of the age 1 recruits estimated from the March survey for each year.
- The number of age 1 recruits for each year (1984–2002) is estimated within the model.
- To enable the age structure of the initial population to be estimated independently of the survey indices there is an additional parameter (*mult*) which can be used to scale the abundance of the initial population with the age structure fixed.
- Selection of the spring survey and commercial catch are described by:

$$S(l, L) = \frac{\delta}{1 + e^{-\alpha - \beta l - \gamma L}} \tag{15}$$

  with $\gamma = 0$, $\delta = 1$ and $\alpha$ and $\beta$ estimated for each fleet separately.
- Selection of immature cod by mature cod is defined by equation 7 with $p_0 = 0$, $p_2 = 1$ and the other parameters estimated within the model.
- As cod growth is not linked to consumption, there is no constraint from growth on the total consumed. If consumption in the wild is taken to be half that of maximum consumption estimated from feeding experiments, then parameter estimates are available. Maximum consumption is defined in equation 11 and can be parameterised from the daily food intake equation from Jobling (1988) to

$$H_P(L, T) = m_0 e^{(0.104T - 0.000112T^3)} L^{2.406}$$

  assuming that $W = 0.01L^3$ with $W$ in grammes and $L$ in cm.

  If the prey is cod, with an energy content of 4.2kJ/g then $m_0 = 1.98 * 10^{-5}$ results in consumption of half the monthly maximum in kilograms. For the *target consumption* $\tau_{P,p,L,l}$ to then be half the maximum possible consumption, the *feeding level* is then required to be $\psi_P(L) = 1$ which is achieved by setting the constant $A_2 = 0$.

# 9 Technical details

As noted by several authors (e.g. Methot (1989)), considerable attention needs to be given to the weights attributed to each negative likelihood component . Weights here are obtained based on the method proposed by Stefansson (1998) & Stefansson (2004). The initial weights are set to the inverse of the negative likelihood function at the initial parameter values, which are arbitrarily chosen. At this stage the total negative likelihood score at the starting point is equal to the total number of components. The specially developed iterative reweighting scheme of Stefansson (2004) is subsequently used for estimation.

Each component is taken in sequence, first the weight of that component is multiplied by 1000, the likelihood function is then minimized and the result for this particular likelihood component is taken as a measure of how well the model can best fit to this data set. That best (minimum) value, scaled to the number of data points (degrees of freedom), is used as a variance estimate, to become an inverse final weight for this component. While the number of terms is easily determinable for some likelihood components, such as survey indices, it is less clear how many should be assigned to data types such as age-length keys for which many values are expected to be zero. In this example, the degrees of freedom for age-length keys were estimated by deciding upon a number

of potentially informative length cells for each year/step/age cell. As there is only one datum for each survey index in each year, the equivalent indices for the spring and autumn surveys are weighted simultaneously.

Ideally the subsequent point estimate would be based on these iteratively chosen weights. Unfortunately, the different likelihood data sources, even with similar objective functions, are not equivalent. In particular, the survey indices are considerably more constrained than the biological sampling data (i.e. length distribution, age length keys and age distributions). For this reason the final parameter estimation is done sequentially with the order of parameter estimation, to a certain extent, model dependent. There is, however a general procedure. Parameters associated with the biological sampling likelihood components are estimated first with the weights on all likelihood components as estimated from the iterative procedure. These parameters (e.g those for growth and the age structure of the initial population) are then fixed and the weights on the likelihood components set to zero. Parameters associated with the survey indices (including the initial population multiplier) are then estimated.

# 10 Results

Parameter estimation and model evaluation are described for a variety of models: a basic cod model with two stocks (immature and mature) and no predation, the same model with predation by mature cod, the basic model with no age disaggregated likelihood data and the basic model with the likelihood data aggregated in 10cm length groups. In each model there are a variety of data components and the emphasis of different datasets can be seen to push the solution in different directions.

## 10.1 Model with no predation: model 1

The minimum sum of squares from the iterative weighting procedure described on page 15 are given in table 1 for the biological sampling and maturity ratio likelihood components and table 2 for the survey index components. Bounds and understocking are zero for all optimised runs and ignored here.

Tables 1 & 2 illustrate the conflicting information from the different data sources. For example, in table 1 the minimum sse from weighting the survey age distribution (ADs) is 0.094 whereas the same component has a score of 0.534 (a value 5.7 times greater) when the catch age distribution is heavily weighted. The information from these data sources is clearly contradictory within the structure of this implementation of Gadget.

The difference in information deriving from the different components is particularly large for the survey indices (table 2). For example the likelihood scores for length group 1 are seven to eleven times greater when length groups 2 or 3 are heavily weighted.

| | mat | LDs | LDc | ALKs | ALKc | ADs | ADc | base | final |
|---|---|---|---|---|---|---|---|---|---|
| mat | **0.131** | 0.36 | 0.403 | 0.274 | 0.336 | 0.229 | 0.325 | 0.307 | 0.294 |
| LDs | 0.612 | **0.046** | 0.126 | 0.18 | 0.142 | 0.164 | 0.122 | 0.17 | 0.07 |
| LDc | 3.745 | 2.744 | **1.184** | 4.2 | 2.629 | 19.13 | 1.895 | 3.372 | 1.646 |
| ALKs | 0.534 | 0.273 | 0.84 | **0.088** | 0.152 | 0.11 | 0.169 | 0.155 | 0.213 |
| ALKc | 7.931 | 5.838 | 15.02 | 3.804 | **2.983** | 13.9 | 3.278 | 4.312 | 5.13 |
| ADs | 2.3 | 0.768 | 1.679 | 0.279 | 0.501 | **0.094** | 0.534 | 0.604 | 0.716 |
| ADc | 29.02 | 13.19 | 49.65 | 14.13 | 8.726 | 55.43 | **7.422** | 15.4 | 13.09 |

Table 1: Minimum sums of squares from iterative optimisations with each column representing a run with that component heavily weighted. The values in bold are the sums of squares of the heavily weighted component in each optimisation. The baseline values are given in the column 'base' and those of the final run in column 'final'. These components are from the biological sampling likelihood components (LD – length distribution, ALK – age length key, AD – age distribution, with s and c indicating whether survey or catch data) and the ratio of immature to mature fish (mat).

| | $I_1$ | $I_2$ | $I_3$ | base | final |
|---|---|---|---|---|---|
| $I_{1S}$ | **1.119** | 12.46 | 11.2 | 3.837 | 3.699 |
| $I_{2S}$ | 3.527 | **0.512** | 6.189 | 0.634 | 0.825 |
| $I_{3S}$ | 1.578 | 1.199 | **0.56** | 1.243 | 0.649 |
| $I_{1A}$ | **0.156** | 1.148 | 1.203 | 0.75 | 0.769 |
| $I_{2A}$ | 2.207 | **0.005** | 3.793 | 0.01 | 0.001 |
| $I_{3A}$ | 1.525 | 1.005 | **0.424** | 0.995 | 1.194 |

Table 2: Minimum sums of squares from iterative optimisations with each column representing a run with that component heavily weighted. The values in bold are the sums of squares of the heavily weighted component in each optimisation – with two indices heavily weighted in each optimisation. The base run values are given in the column 'base' and those of the final run in the last column. These components are survey indices where $I_i$, $i = 1, 2, 3$ are the length groups and the S, A subscripts indicate the spring and autumn surveys.

The weights determined from the iterative procedure are shown in table 3 along with the ratio of the sum of squares from the final model to the minimum for each component (tables 1 & 2). As a result of incorporating information from the different data sources, the final sse for each component is greater than the minimum from the iterative weighting. For length group 2 of the survey indices, the total score is greater although the score for the autumn index alone is lower in the final model. The ratio of final see to minimum sse suggests that relatively more information is retained from the catch than the survey data (table 3).

Parameter estimation using the weights in table 3 was done in three stages. 1) all likelihood data were used, 2) growth, selection (catch and survey) and initial population parameters were fixed, the LD, ALK and AD weights were set to zero and the initial population multiplier was used 3) maturity parameters fixed and the maturity likelihood data to zero. The parameters estimated from the procedure are shown figure 4.

Plots of the fitted log-linear regression are shown in figure 5. The slopes for length group 1 are 1.64 and 0.54 for the spring and autumn surveys respectively and for length groups 2 the slopes are 1.62 and 1.83. The highest weight of the survey indices is for length group 2 of the autumn survey (table 2) and consequently the modelled population fits this component particularly well. For the survey indices, the largest decreases between the likelihood score from the weighted run and the final run are for length groups 1 and 3 of the autumn survey which can be seen to have the worst fit to the modelled population (figure 5).

| Component | Weight | $sse_f/sse_m$ |
|---|---|---|
| mat | 6032 | 2.24 |
| LDs | 26333 | 1.52 |
| LDc | 10358 | 1.39 |
| ALKs | 34200 | 2.42 |
| ALKc | 8335 | 1.72 |
| ADs | 2307 | 7.62 |
| ADc | 236 | 1.76 |
| $I_{1S}$ | 16.09 | 3.31 |
| $I_{2S}$ | 35.18 | 1.61 |
| $I_{3S}$ | 32.13 | 1.16 |
| $I_{1A}$ | 51.35 | 4.93 |
| $I_{2A}$ | 1514 | 0.20 |
| $I_{3A}$ | 18.87 | 2.82 |

Table 3: Component weights calculated from the minimum sums of squares and $sse_f/sse_m$, the ratio of the sums of squares from the final model ($sse_f$) with the minimum sse ($sse_m$) from the iterative weighting.

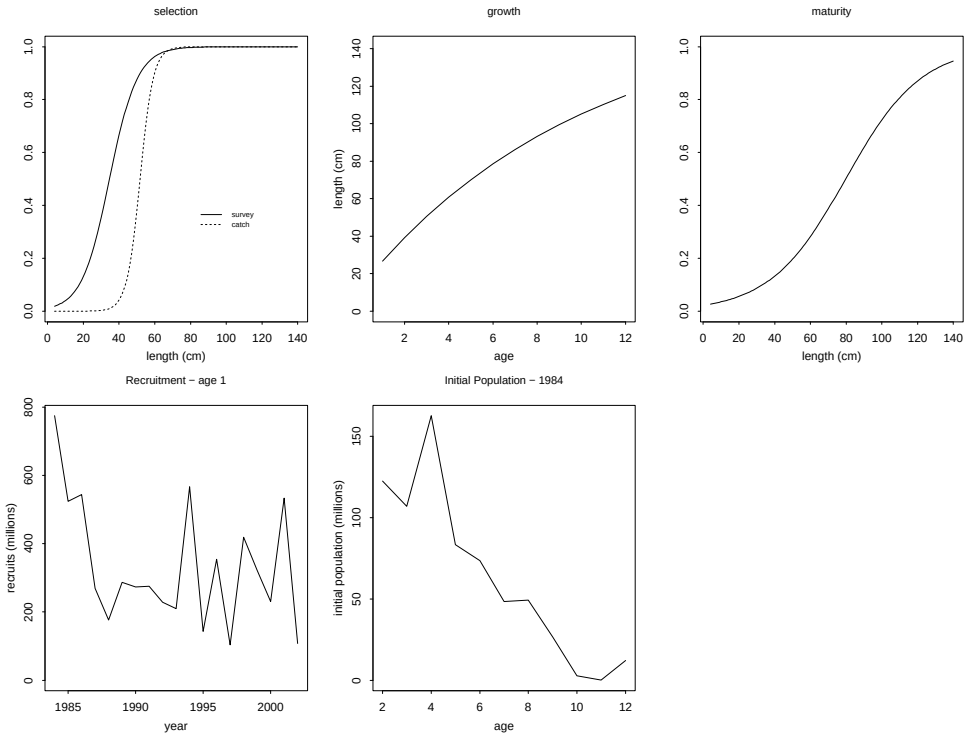



Figure 4: Parameter estimates from the final optimisation: selection patterns of the spring survey and catch, growth, maturation, number of age 1 recruiting into the model and the initial population age 2–12

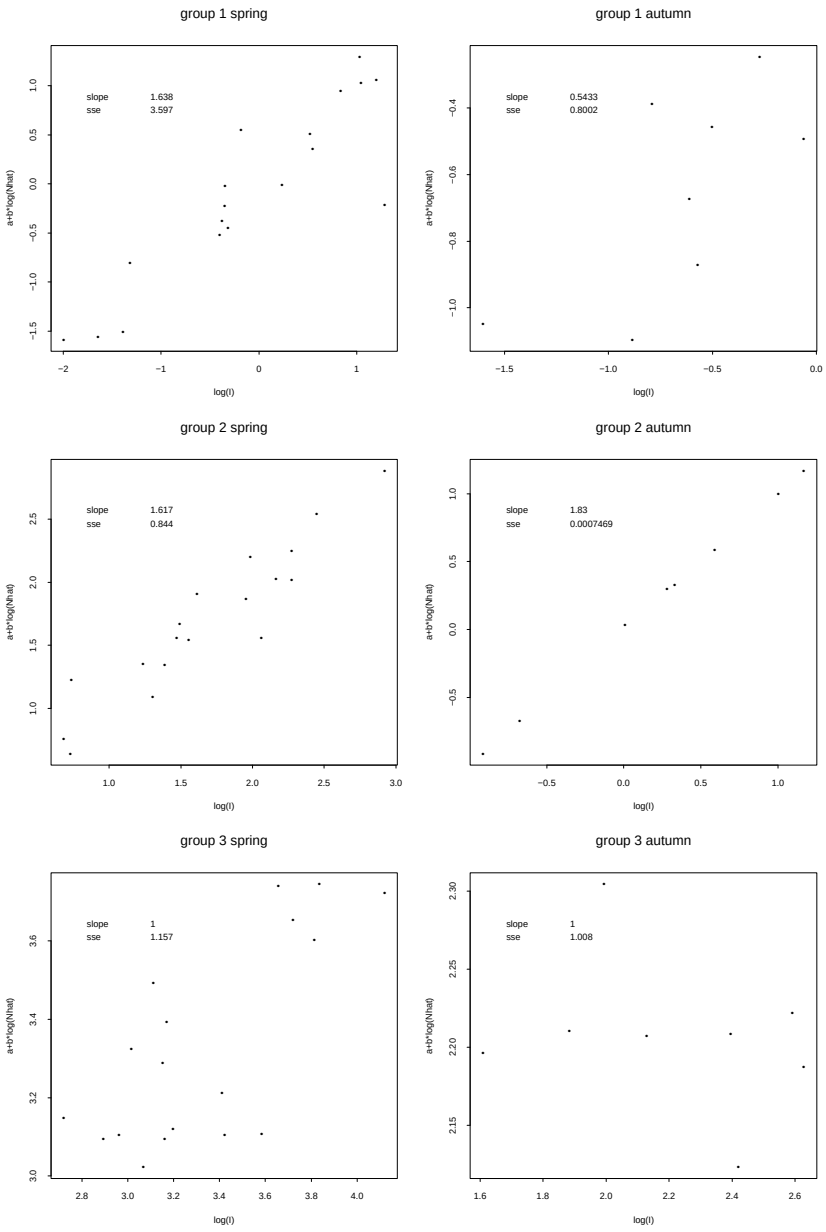



Figure 5: Plots of the log-linear regression as estimated in Gadget, with one plot for each likelihood component and the corresponding slope (estimated for length groups 1 and 2) and sse for each survey index. log(I) is the index and a+b*log(N) the regression fitted to the modelled population.

## 10.2 Model with predation: model 2

The minimum sums of squares from the iterative weighting procedure are given in table 4 for the biological sampling and maturity ratio components and table 5 for the survey indices. In addition, table 6 contains the minimum sums of squares for the stomach data likelihood components. Bounds and understocking are zero for all optimised runs. The similarities in the values from model 1 and model 2 (comparing tables 1 & 4 and tables 2 and 5) indicates that the parameter settings for the optimisation routines and the results of the optimisations are adequate.

| | mat | LDs | LDc | ALKs | ALKc | ADs | ADc | base | final |
|---|---|---|---|---|---|---|---|---|---|
| mat | **0.085** | 0.472 | 0.462 | 0.209 | 0.328 | 0.963 | 0.358 | 0.222 | 0.235 |
| LDs | 0.3729 | **0.037** | 0.139 | 0.168 | 0.126 | 0.148 | 0.124 | 0.083 | 0.07 |
| LDc | 10.3 | 6.315 | **1.087** | 6.502 | 2.742 | 16.99 | 2.516 | 1.514 | 1.523 |
| ALKs | 0.2219 | 0.246 | 0.611 | **0.081** | 0.162 | 0.191 | 0.2 | 0.141 | 0.211 |
| ALKc | 8.733 | 7.001 | 12.06 | 5.033 | **2.98** | 10.32 | 3.123 | 3.762 | 5.121 |
| ADs | 1.113 | 0.687 | 1.25 | 0.226 | 0.546 | **0.107** | 0.615 | 0.397 | 0.717 |
| ADc | 40.44 | 21.76 | 28.19 | 17.35 | 8.718 | 19.97 | **7.339** | 8.508 | 12.7 |

Table 4: Minimum sums of squares from iterative optimisations with each column representing a run with that component heavily weighted. The values in bold are the sums of squares of the heavily weighted component in each optimisation. The base run values are given in the column 'base' and those of the final run in the last column. These components are from the biological sampling likelihood components and the ratio of immature to mature fish. Row and column labels are identical to those in table 1.

| | $I_1$ | $I_2$ | $I_3$ | base | final |
|---|---|---|---|---|---|
| $I_{1S}$ | **1.262** | 13.85 | 15.97 | 2.541 | 3.756 |
| $I_{2S}$ | 4.726 | **0.529** | 4.51 | 1.409 | 0.797 |
| $I_{3S}$ | 1.935 | 2.046 | **0.564** | 1.27 | 0.636 |
| $I_{1A}$ | **0.102** | 1.316 | 1.501 | 0.557 | 0.757 |
| $I_{2A}$ | 2.768 | **0.008** | 3.793 | 0.233 | 0.002 |
| $I_{3A}$ | 1.3 | 1.249 | **0.2** | 1.023 | 1.183 |

Table 5: Minimum sums of squares from iterative optimisations with each column representing a run with that component heavily weighted. The values in bold are the sums of squares of the heavily weighted component in each optimisation. The base run values are given in the column 'base' and those of the final run in the last column. These components are survey indices. Row and column labels are identical to those in table 2.

The addition of predation into the model involves the inclusion of relatively sparse data. While data are available on the proportion of cod in the diet of cod, the low proportion means that length sampling of cod prey is limited and the data sparse. Further aggregation of the data either by predator length group or prey length group could lead to little information on the optimal predator length:prey length relationship or the shape of the selection curve. The sparseness of the likelihood data means that the difference between the likelihood score when heavily weighting the prey length distributions, PLs and PLo, compared to weighting the other components is much less than for other data types (table 6).

| | PBs | PLs | PBo | PLo | base | final |
|---|---|---|---|---|---|---|
| PBs | **0.003** | 0.025 | 0.016 | 0.023 | 0.017 | 0.014 |
| PLs | 87.28 | **65.46** | 95.59 | 67.8 | 70.89 | 82.55 |
| PBo | 0.181 | 0.193 | **0.154** | 0.192 | 0.175 | 0.171 |
| PLo | 79.58 | 68.23 | 88.29 | **65.81** | 71.07 | 76.85 |

Table 6: Minimum sums of squares from iterative optimisations with each column representing a run with that component heavily weighted. The values in bold are the sums of squares of the heavily weighted component in each optimisation. The base run values are given in the column 'base' and those of the final run in the last column. These components are stomach content data.

Figure 6 illustrates the difference in predicted recruitment when emphasis is put on the different survey length groups. While there are similarities e.g. all plots have a peak in 1994 followed by a low value in 1995, there are many differences between the estimated values. The correlations between the recruit series are:

| Indices | cor |
|---|---|
| $I_1$ and $I_2$ | 0.28 |
| $I_1$ and $I_3$ | 0.60 |
| $I_2$ and $I_3$ | 0.52 |

It should be noted that length group 1 provides no information for 1984 as the first survey index is for 1985.

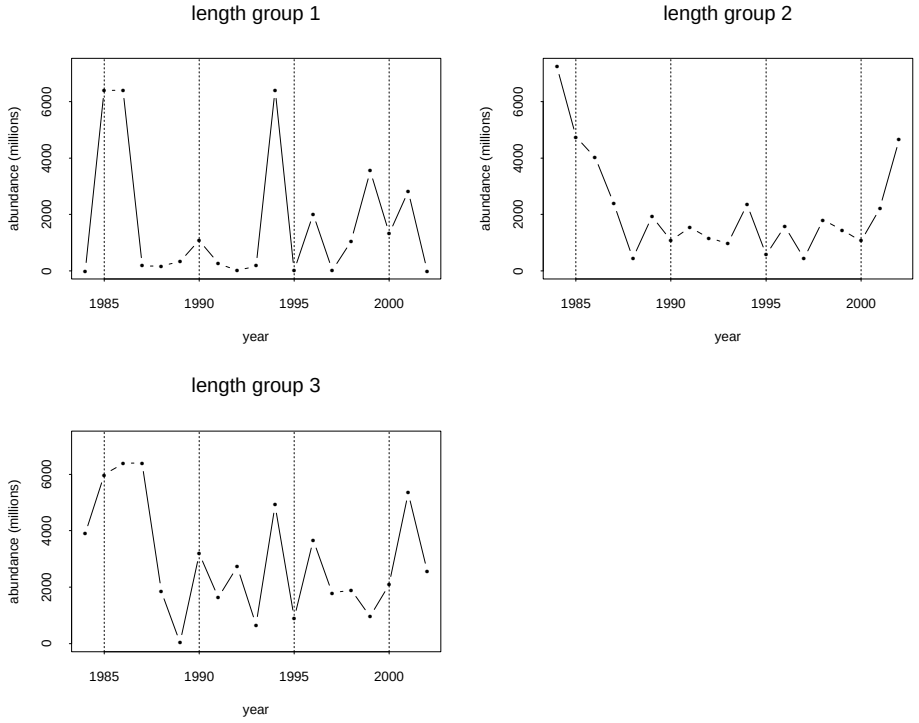



Figure 6: Recruitment estimated from the iterative weighting procedure. The plots show recruitment estimated with the survey data for length group 1, length group 2 and length group 3 heavily weighted respectively.

As parameter estimation using the prey biomass components (to estimation consumption of other food) needs to be done at the same time as estimation using the survey indices, which estimate

the total abundance, the prey biomass data from the survey (PBs) were not included in the later stages of the estimation procedure as the weight was so large (table 7). The spring survey stomach content biomass ratios are almost the same in every year resulting in a low minimum sse. The sparse data for the prey length distributions, however, result in low weights for these components.

The survey indices with the least difference between the minimum sse and the sse with the final parameters are length groups 2 for both surveys and length group 3 for the spring survey (table 7). This is reflected in the fit of the indices to the modelled abundance (figure 8).

| Component | weight | $sse_f/sse_m$ |
|---|---|---|
| mat | 9364 | 2.76 |
| LDs | 32992 | 1.89 |
| LDc | 11283 | 1.40 |
| ALKs | 37200 | 2.60 |
| ALKc | 8344 | 1.72 |
| ADs | 2019 | 6.70 |
| ADc | 239 | 1.73 |
| $I_{1S}$ | 14.26 | 2.98 |
| $I_{2S}$ | 34.03 | 1.51 |
| $I_{3S}$ | 31.90 | 1.13 |
| $I_{1A}$ | 78.74 | 7.42 |
| $I_{2A}$ | 1062 | 0.25 |
| $I_{3A}$ | 39.98 | 5.92 |
| PBs | 10529 | 4.67 |
| PLDs | 3.71 | 1.26 |
| PBo | 507.81 | 1.11 |
| PLo | 3.42 | 1.17 |

Table 7: Component weights calculated from the minimum sums of squares and $sse_f/sse_m$, the ratio of the sums of squares from the final model ($sse_f$) with the minimum sse ($sse_m$) from the iterative weighting.

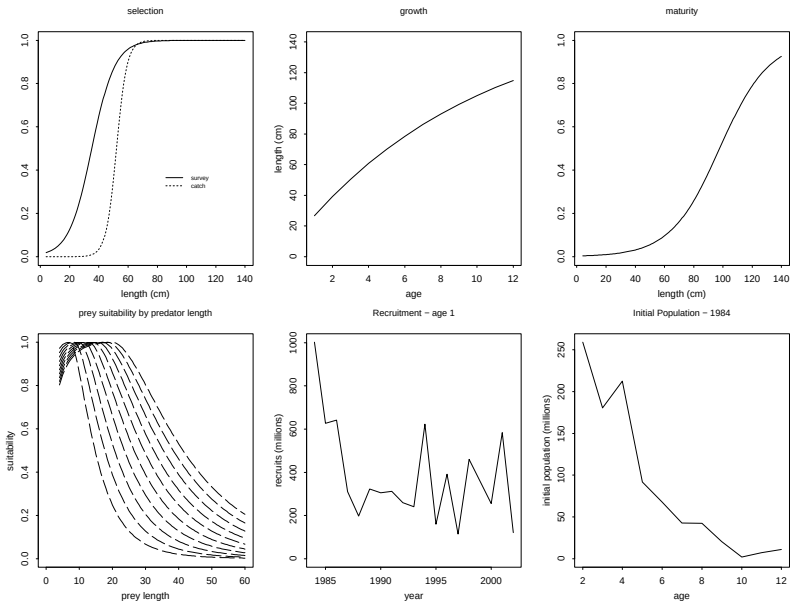



Figure 7: Parameter estimates from the final optimisation: selection patterns of the spring survey and catch, growth, maturation, selection curve of cod prey by cod predators for a selection of predator lengths, the number of age 1 recruiting into the model and the initial population age 2–12

Plots of the fitted log-linear regression are shown in figure 8. The slopes for length group 1 are 1.66 and 0.59 for the spring and autumn surveys respectively and for length groups 2 the slopes are 1.62 and 1.86. These estimated slopes are similar to those for model 1. The low consumption of cod by cod in Icelandic waters is not expected to have much impact on the population dynamics and there is no evidence here that it does.

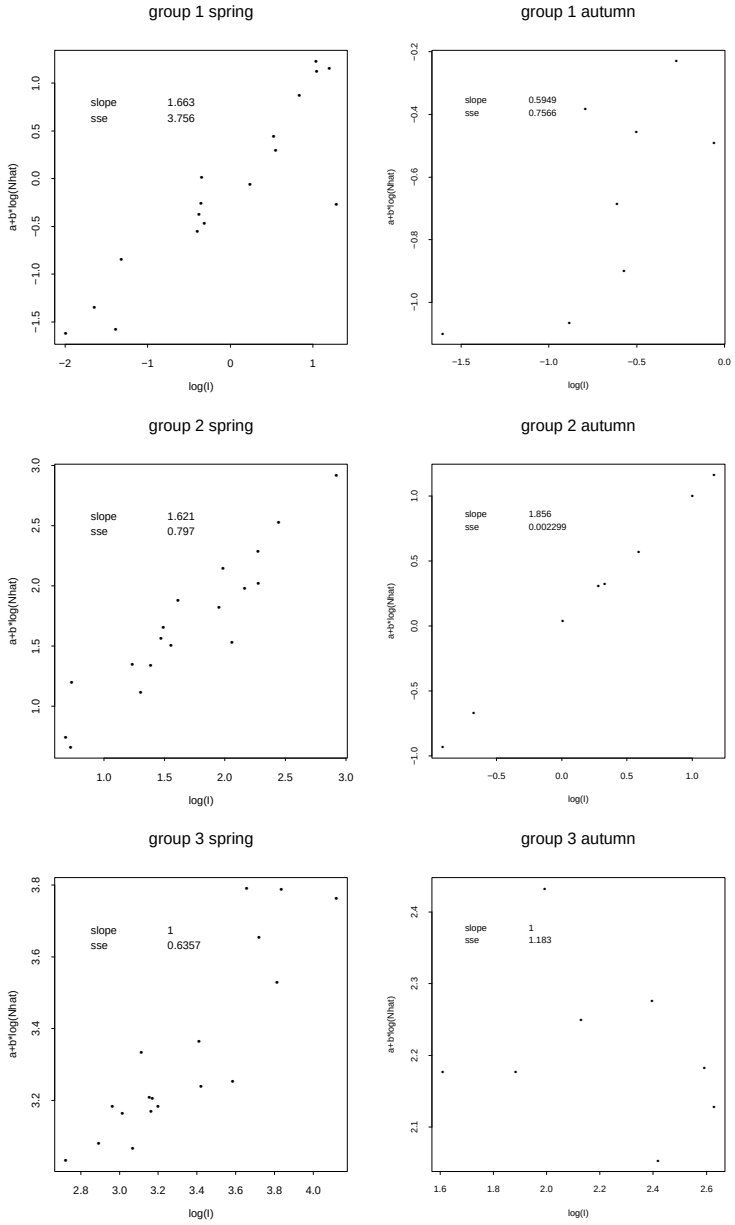



Figure 8: Plots of the log-linear regression as estimated in Gadget, with one plot for each likelihood component and the corresponding slope (estimated for length groups 1 and 2) and sse for each survey index. log(I) is the index and a+b*log(N) the fitted regression to the modelled population.

### 10.2.1 Residuals

The fit of the model to the likelihood data can be considered in more detail by considering residuals from the fit of length distribution components and modelled catch length distributions calculated as $p_{y,s,l} - \pi_{y,s,l}$ where $p_{y,s,l}$ is the observed length distribution as proportions by year and step and $\pi_{y,s,l}$ is the modelled catch length distribution as proportions by year and step.

The patterns in the residuals from the survey length distribution likelihood component and the modelled survey catch (figures 9 & 10) and equivalent residuals for the catch length distribution component and modelled commercial catch (figure 11) indicate that growth in this implementation of Gadget may be insufficiently flexible. It is, however, to be expected that the length distributions from the modelled population will be smoother than those observed as samples are affected by intra-haul correlation (Pennington and Volstad 1994) with fish of a similar length being caught together (Hrafnkelsson and Stefansson 2002).

A model with growth affected by consumption might increase the flexibility of growth, as might variable growth rates for different years or time periods. For the catch data, disaggregating the commercial fleet into different gear types, which apart from having different selection patterns also operate during different seasons, might also improve the model.

Commercial catch residuals from January 1985 are considerably greater than those from other timesteps (figure 11) and possibly should be removed from the model.

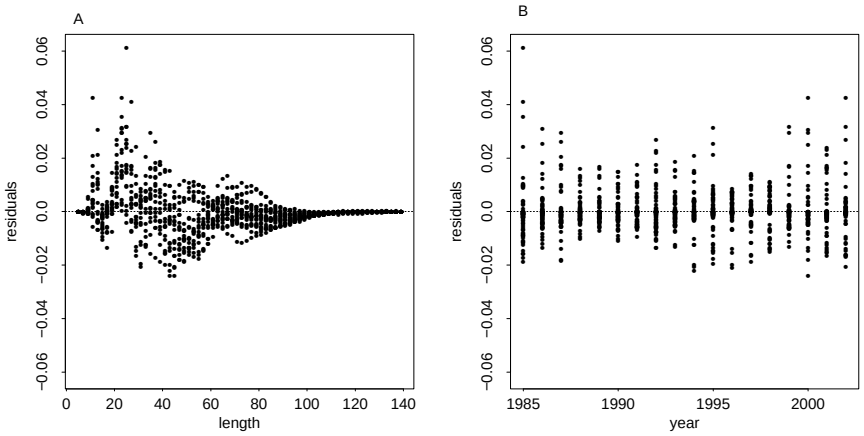



Figure 9: Length distribution residuals from the spring survey. A) by 2cm length group and B) by year.

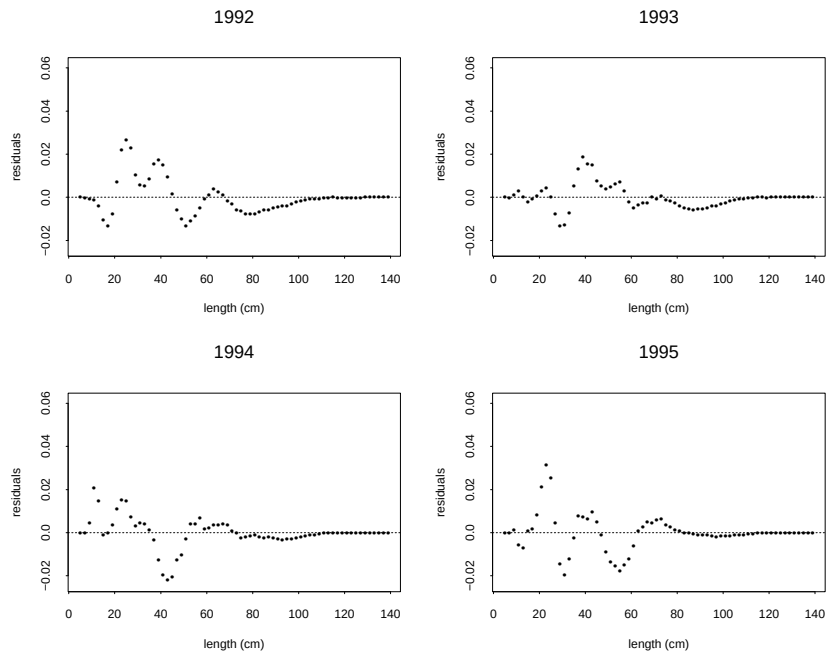



Figure 10: Length distribution residuals from the spring survey for an arbitrary selection of years 1992–1995

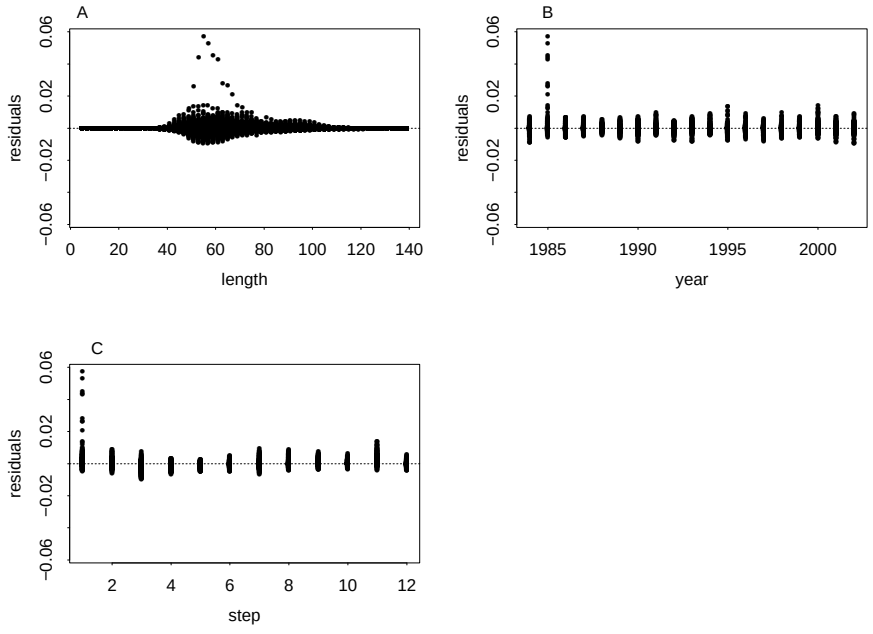



Figure 11: Length distribution residuals from the catch data. A) by 2cm length group, B) by year and C) by month.

The total likelihood score by year, step and component can also be plotted as in figures 12 for the survey data and figures 13 & 14 for the catch data. The poorer fit for the maturity data for the initial years may be affected by the initial distribution of fish between the immature and mature stock components. As both the maturity and age-length keys components (and to a lesser extent the age distribution data) have relatively high scores for 1985–1989 it is more likely that the cause is the otolith sampling protocol for the survey which was subsequently changed.

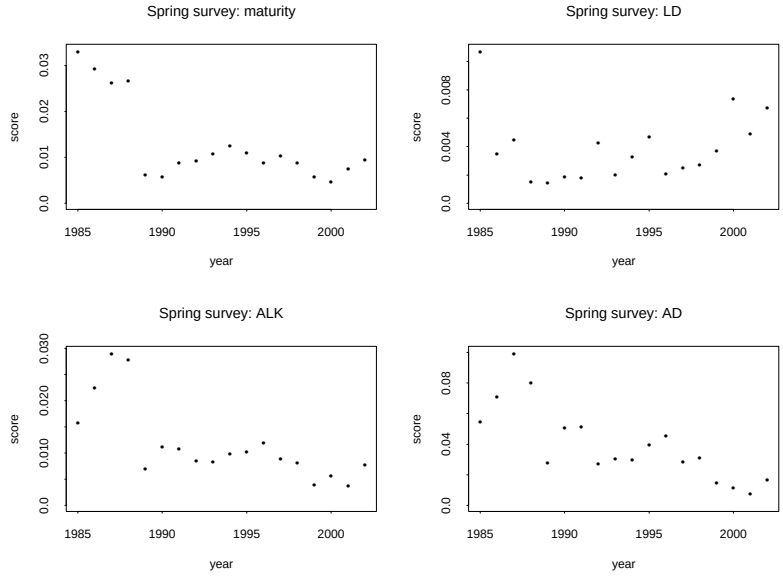



Figure 12: Likelihood scores by year for the spring survey components: maturity, length distribution, age length keys and age distribution.

It can be seen in figure 13 that the likelihood data do not fit all years equally, with higher likelihood scores for the years 1995–1999 for all biological sampling datasets. There is also a seasonal pattern (figure 14) with higher and more variable scores in the spring. Disaggregating the commercial fleet by gear type might reduce some of the seasonality in the likelihood scores. Including a separate gill net fleet is of particular interest as gill net use is seasonal and used to target larger spawning fish. The sampling of the gill net fishery during the spawning season is also disproportionately large in relation to landings.

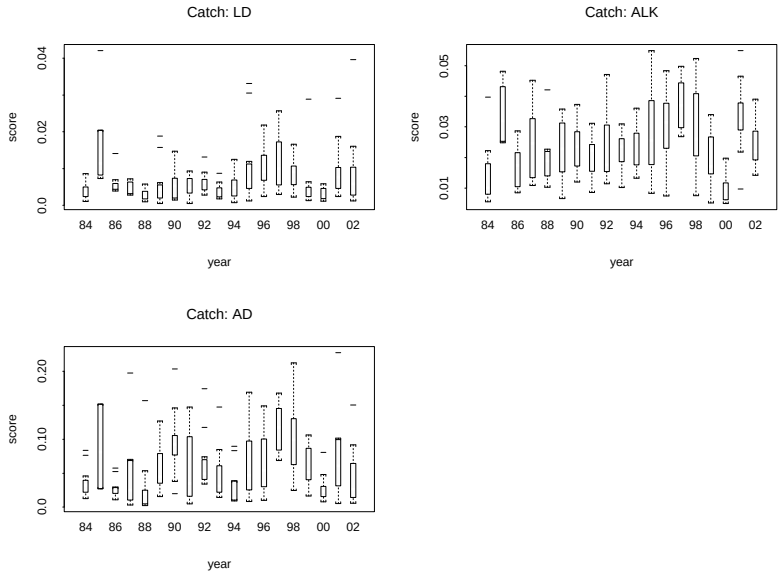



Figure 13: Boxplots of the likelihood scores by year for the commercial catch components: length distribution, age length keys and age distribution.

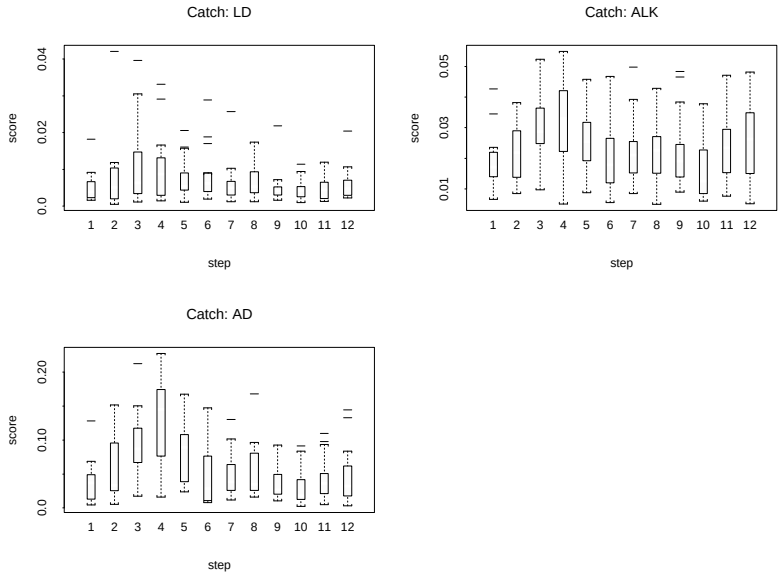



Figure 14: Boxplot of the likelihood scores by month for the commercial catch components: length distribution, age length keys and age distribution.

## 10.3 Model with no age data: model 3

For model 3, model 1 is repeated with no age data likelihood components. The remaining likelihood components are: length distributions for the spring survey and catch, the ratio of immature to mature by length group and the survey indices by length groups. Apart from the growth function being fixed and different initial parameters the procedure is the same as with age data.

The minimum sums of squares from the iterative weighting procedure are in table 8 for all components.

| | mat | LDs | LDc | $I_1$ | $I_2$ | $I_3$ | base | final |
|---|---|---|---|---|---|---|---|---|
| mat | **0.136** | 0.363 | 0.39 | 1.086 | 0.565 | 1.2 | 0.348 | 0.283 |
| LDs | 0.532 | **0.037** | 0.082 | 0.484 | 0.268 | 0.503 | 0.155 | 0.064 |
| LDc | 11.11 | 3.008 | **1.625** | 12.77 | 7.216 | 8.815 | 3.752 | 1.409 |
| $I_{1S}$ | 10.63 | 3.268 | 14.9 | **1.438** | 12.73 | 11.94 | 5.271 | 3.694 |
| $I_{2S}$ | 6.685 | 1.402 | 6.685 | 2.863 | **0.331** | 6.204 | 0.34 | 0.523 |
| $I_{3S}$ | 5.345 | 0.98 | 14.23 | 1.791 | 1.929 | **0.56** | 1.148 | 0.651 |
| $I_{1A}$ | 0.918 | 0.642 | 0.806 | **0.162** | 1.028 | 1.202 | 0.592 | 0.646 |
| $I_{2A}$ | 1.874 | 0.344 | 0.45 | 1.588 | **0.058** | 3.793 | 0.075 | 0.071 |
| $I_{3A}$ | 1.722 | 0.964 | 1.441 | 1.333 | 0.967 | **0.418** | 0.998 | 1.165 |

Table 8: Minimum sums of squares from iterative optimisations with each column representing a run with that component heavily weighted. The values in bold are the sums of squares of the heavily weighted component in each optimisation. The base run values are given in the column 'base' and those of the final run in the last column. These components are from the biological sampling likelihood components and the ratio of immature to mature fish. Row and column labels are identical to those in table 1.

Table 9 contains the weights calculated from the iterative procedure and the ratio of the sum of squares from the final model to the minimum for each component. In most cases, the difference between the minimum and final sse is less than for models 1 and 2 as there are fewer data sets providing conflicting information. The final parameter estimates are shown in figure 15.

| Component | weight | $sse_f/sse_m$ |
|---|---|---|
| mat | 5836 | 2.08 |
| LDs | 32701 | 1.73 |
| LDc | 7548 | 0.87 |
| $I_{1S}$ | 12.52 | 2.57 |
| $I_{2S}$ | 54.41 | 1.58 |
| $I_{3S}$ | 32.15 | 1.16 |
| $I_{1A}$ | 49.44 | 3.99 |
| $I_{2A}$ | 137.55 | 1.22 |
| $I_{3A}$ | 19.12 | 2.79 |

Table 9: Component weights calculated from the minimum sums of squares and $sse_f/sse_m$, the ratio of the sums of squares from the final model ($sse_f$) with the minimum sse ($sse_m$) from the iterative weighting.

Plots of the fitted log-linear regression are shown in figure 16. The slopes for length group 1 are 1.61 and 0.58 for the spring and autumn surveys respectively and for length groups 2 the slopes are 1.64 and 1.70. As before the highest weight of the survey indices is for length group 2 of the autumn survey (table 8) but the weight is considerably lower than for models 1 and 2. The sse from the log-linear regression is correspondingly higher at 0.07 for model 3 compared to 0.0007 and 0.002 for models 2 and 3 respectively.

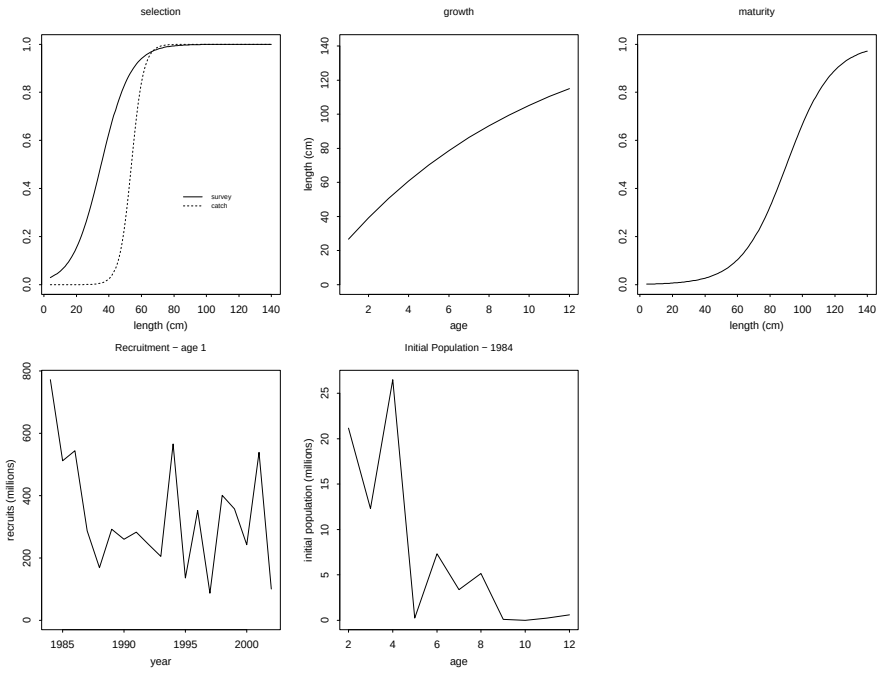



Figure 15: Parameter estimates from the final optimisation: selection patterns of the spring survey and catch, growth (fixed), maturation, number of age 1 recruiting into the model and the initial population age 2–12

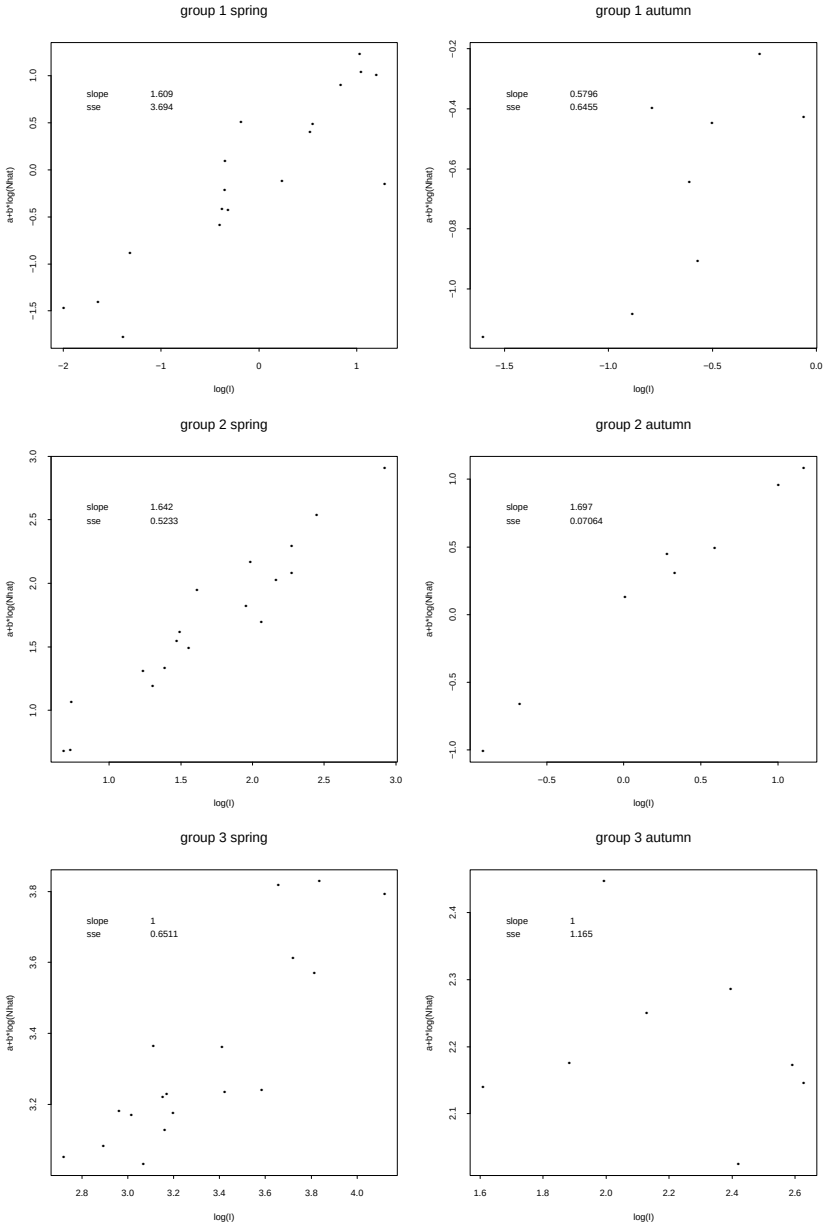



Figure 16: Plots of the log linear regression as estimated in Gadget, with one plot for each likelihood component and the corresponding slope (estimated for length groups 1 and 2) and sse for each survey index. log(I) is the index and a+b*log(N) the fitted regression to the modelled population.

## 10.4 Model with aggregated likelihood components: model 4

Model 4 is equivalent to model 1 with the length distribution, age-length key and maturity ratio likelihood components aggregated in 10cm length groups. Survey indices are identical to those used for the other models. All parameters are estimated as for model 1.

Tables 10 & 11 contain the minimum sum of squares from the iterative weighting procedure. The minima for the length data in table 10 are greater (by as much as a factor of 4) than those for model 1.

| | mat | LDs | LDc | ALKs | ALKc | ADs | ADc | base | final |
|---|---|---|---|---|---|---|---|---|---|
| mat | **0.371** | 1.803 | 1.485 | 0.811 | 1.358 | 2.081 | 1.154 | 1.246 | 1.407 |
| LDs | 2.442 | **0.203** | 0.370 | 0.739 | 0.496 | 0.557 | 0.359 | 0.697 | 0.331 |
| LDc | 17.240 | 8.465 | **4.621** | 48.920 | 11.440 | 40.010 | 5.807 | 14.760 | 6.059 |
| ALKs | 1.292 | 1.796 | 1.721 | **0.136** | 0.309 | 1.119 | 0.377 | 0.334 | 0.397 |
| ALKc | 21.720 | 34.390 | 34.210 | 17.180 | **6.258** | 33.350 | 8.418 | 9.747 | 11.200 |
| ADs | 2.609 | 1.609 | 1.485 | 0.149 | 0.506 | **0.103** | 0.489 | 0.614 | 0.625 |
| ADc | 30.540 | 43.240 | 50.810 | 31.310 | 8.927 | 51.350 | **7.375** | 16.850 | 14.660 |

Table 10: Minimum sums of squares from iterative optimisations with each column representing a run with that component heavily weighted. The values in bold are the sums of squares of the heavily weighted component in each optimisation. The base run values are given in the column 'base' and those of the final run in the last column. These components are from the biological sampling likelihood components and the ratio of immature to mature fish. Row and column labels are identical to those in table 1.

The minimum sums of squares for the indices are similar to those for model 3 which was started with similar initial parameter values. Initial parameter values can affect the weighting of the survey indices as they influence the weight of the baseline run.

| | $I_1$ | $I_2$ | $I_3$ | base | final |
|---|---|---|---|---|---|
| $I_{1S}$ | **1.463** | 12.520 | 11.95 | 4.726 | 3.835 |
| $I_{2S}$ | 3.519 | **0.334** | 6.187 | 0.355 | 0.525 |
| $I_{3S}$ | 2.155 | 1.167 | **0.516** | 1.230 | 0.638 |
| $I_{1A}$ | **0.136** | 1.061 | 1.238 | 0.624 | 0.628 |
| $I_{2A}$ | 3.640 | **0.055** | 3.793 | 0.078 | 0.070 |
| $I_{3A}$ | 1.677 | 1.005 | **0.449** | 0.994 | 1.183 |

Table 11: Minimum sums of squares from iterative optimisations with each column representing a run with that component heavily weighted. The values in bold are the sums of squares of the heavily weighted component in each optimisation – with two indices heavily weighted in each optimisation. The base run values are given in the column 'base' and those of the final run in the last column. These components are survey indices where $I_i$, $i = 1, 2, 3$ are the length groups and the S, A subscripts indicate the spring and autumn surveys.

Table 12 contains the weights calculated from the iterative procedure and the ratio of the sum of squares from the final model to the minimum for each component. As indicated by the minimum sums of squares the weighting of the survey indices is similar to that for model 3. In most cases, the ratio is greater for model 4 then for the other models, indicating that the final model fit is poorer to these aggregated likelihood datasets.

The final parameter estimates are shown in figure 17 where it can be seen that the parameter estimates are, in most cases similar to those from the earlier models.

Plots of the fitted log-linear regression are shown in figure 18. The slopes for length group 1 are

| Component | Weight | $sse_f/sse_m$ |
|---|---|---|
| mat | 436.187 | 3.79 |
| LDs | 1238.938 | 1.63 |
| LDc | 690.760 | 1.31 |
| ALKs | 11134.020 | 2.92 |
| ALKc | 2040.213 | 1.79 |
| ADs | 2103.213 | 6.09 |
| ADc | 247.322 | 1.99 |
| $I_{1S}$ | 12.303 | 2.62 |
| $I_{2S}$ | 53.957 | 1.24 |
| $I_{3S}$ | 34.863 | 1.28 |
| $I_{1A}$ | 58.737 | 1.57 |
| $I_{2A}$ | 146.065 | 4.61 |
| $I_{3A}$ | 17.833 | 2.64 |

Table 12: Component weights calculated from the minimum sums of squares and $sse_f/sse_m$, the ratio of the sums of squares from the final model ($sse_f$) with the minimum sse ($sse_m$) from the iterative weighting.

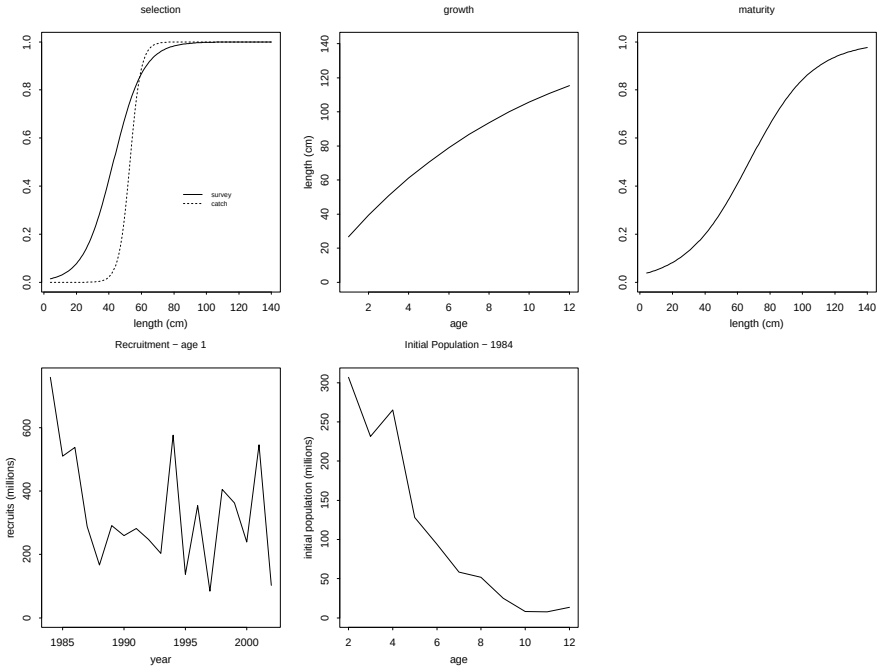



Figure 17: Parameter estimates from the final optimisation: selection patterns of the spring survey and catch, growth (fixed), maturation, number of age 1 recruiting into the model and the initial population age 2–12

1.59 and 0.58 for the spring and autumn surveys respectively and for length groups 2 the slopes are 1.63 and 1.67. Despite being considerably lower than for models 1 and 2, the highest weight of the survey indices is still for length group 2 of the autumn survey (table 12). The sse from the log-linear regression is correspondingly higher at 0.07 for model 4 compared to 0.0007 and 0.002 for models 1 and 2 respectively. In general, the sse for the indices are closer to those for model 3 than models 1 and 2, indicating the effect of the initial parameters on the resulting weights and

fit. The slopes and intercepts are, however, similar for all models.

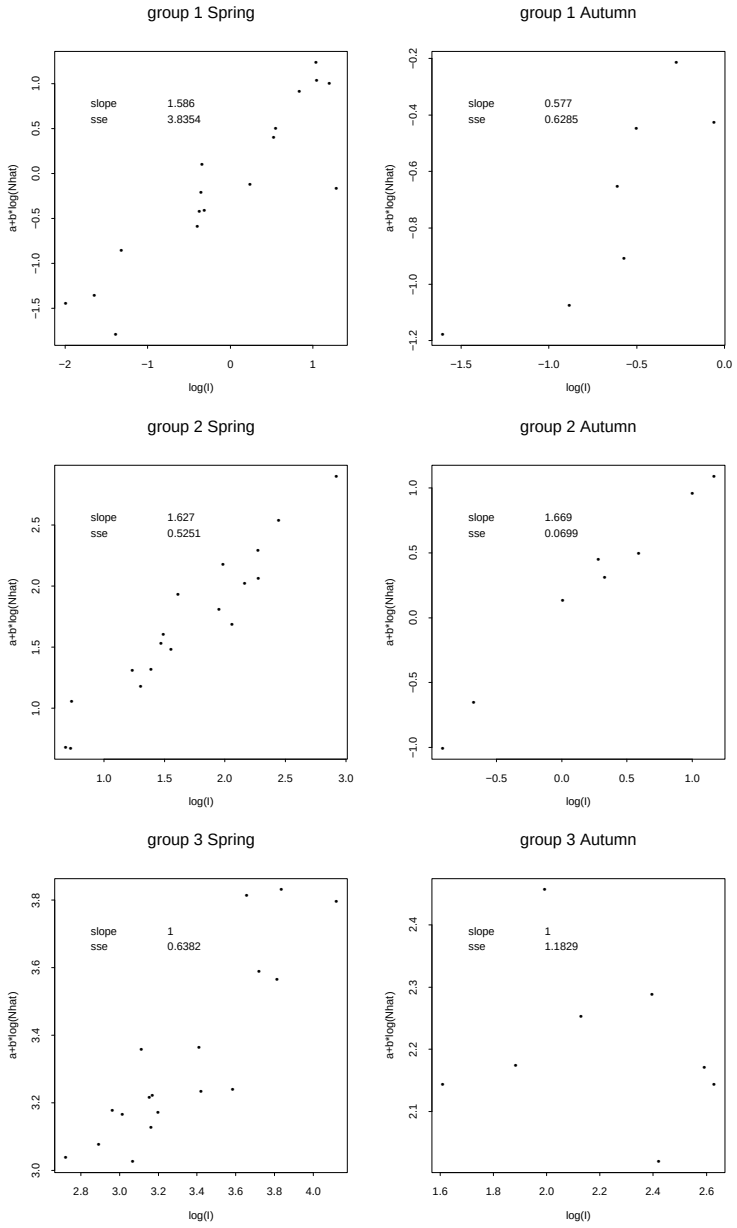



Figure 18: Plots of the log-linear regression as estimated in Gadget for model 4, with one plot for each likelihood component and the corresponding slope (estimated for length groups 1 and 2) and sse for each survey index. log(I) is the index and a+b*log(N) the fitted regression to the modelled population.

## 10.5 Model comparison

The tables summarising the iterative weighting procedures for the four models (tables 1, 2, 3, 4, 5, 6, 8, 9, 10, 11 & 12) along with the plots of the estimated parameter values (figures 4, 7, 15 & 17) indicate consistency in both optimisation and parameter estimation for the first three models despite the differences between the models and initial parameter values.

A comparison of the estimation of the selection pattern and growth (for the models with age data) is shown in figure 19. Estimation of both growth and the selection pattern of the commercial fleet are similar but the survey selection differs when estimated form the aggregated data in model 4. From the length distributions in figure 20 it can be seen that aggregating the survey index loses more information than aggregating the catch data due to the range of lengths in the datasets.

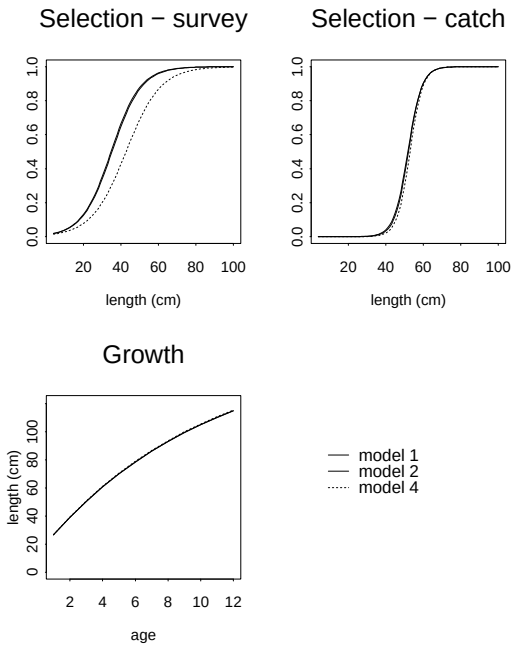



Figure 19: Comparison of selection and growth estimated by the models with age data.

The consistency of the estimated parameters is illustrated further in figure 21 where the estimated recruitment and initial population are compared directly for all the models. There is slightly more recruitment estimated from model 2, which is to be expected with predation, but otherwise recruitment parameters are similar. There is greater variability in the estimates of the initial population, particularly for model 3 with no age data. Most importantly, the stock biomass predicted by each of the models is similar, as shown in figure 22. Differences in the initial population result in different levels of biomass for the first few years, with the trajectory similar for models 1 & 2 and models 3 & 4. This is likely to be a result of the different weights assigned to the survey indices in these models. Differences in spawning stock biomass result from slightly different maturation ogives being estimated.

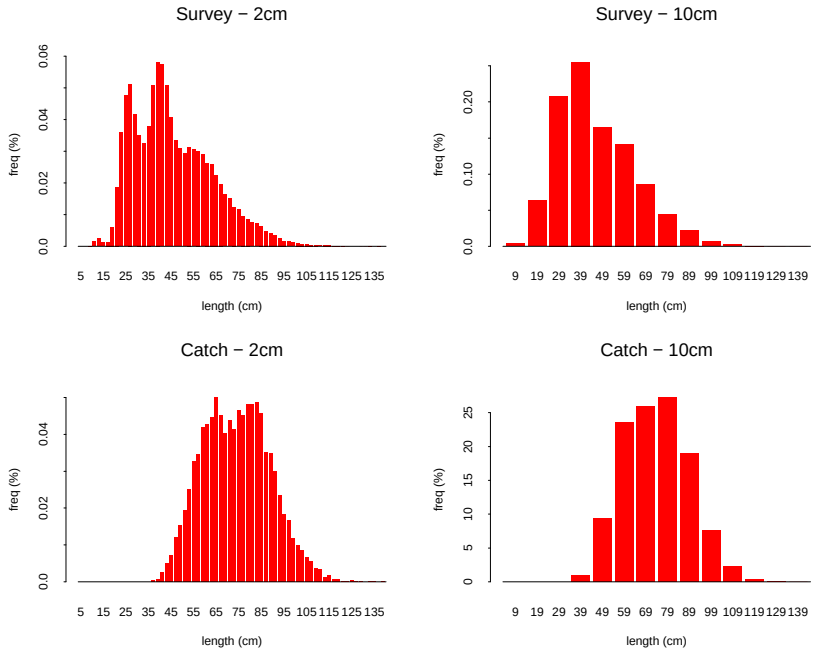



Figure 20: Length distribution from March 1990 for the survey and commercial catch aggregated on 2cm and 10cm.

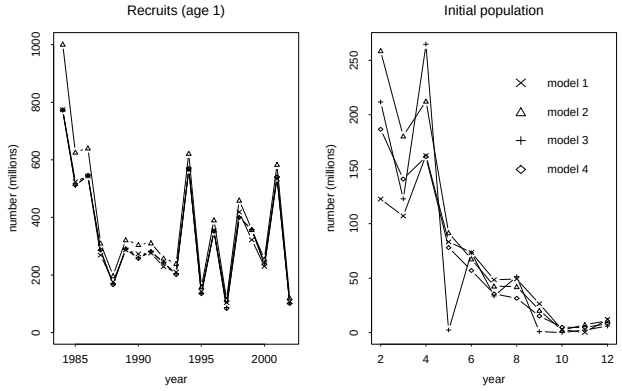



Figure 21: Comparison of the number of recruits (age 1) and the initial population (age 2–12) estimated by the four models.

# 11 Discussion

In this paper it has been demonstrated that it is possible to parameterise a statistical fisheries model using a formal statistical methodology. In these examples more data sources (seventeen likelihood datasets in model 2) and biological processes have been combined in such a model than before. Although these are single species models, it should be noted that there are two stock components which, in the case of model 2, interact. It was found that while it is possible to estimate a considerable number of parameters, it is necessary to fix some.

These examples are data-rich but with judicious use of available data the methodology also seems

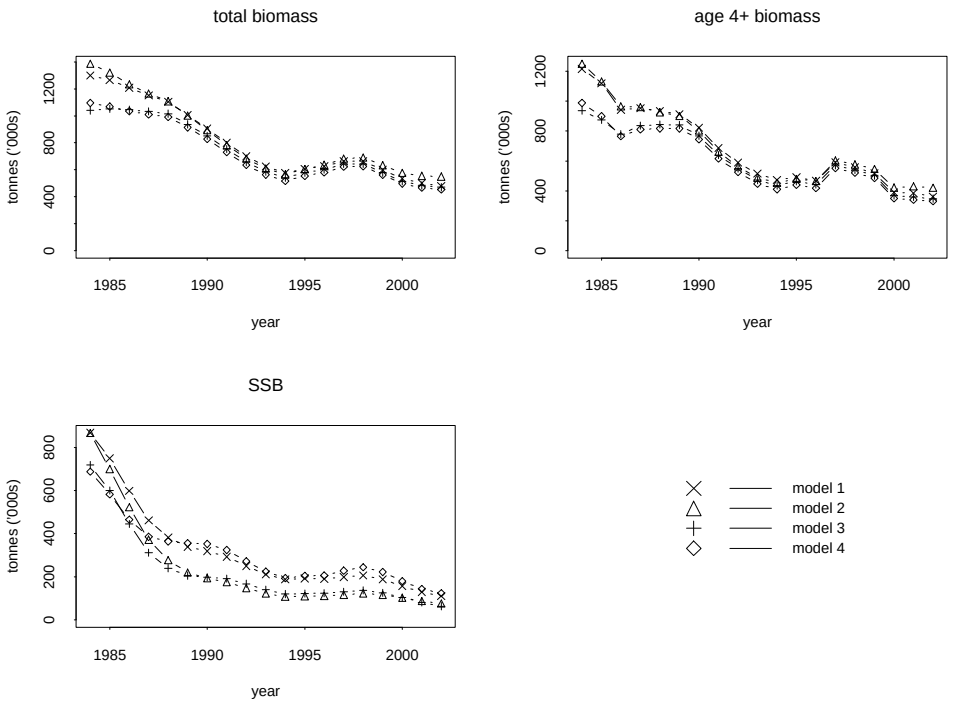



Figure 22: Comparison of the biomass estimated by the four models, for: total (age 1+) biomass, age 4+ biomass and spawning stock biomass.

promising for less data-rich environments. As demonstrated in model 3, with fixed growth the lack of age data does not adversely affect estimates of recruitment or stock biomass. In the case of more aggregated likelihood components, parameter estimation in most cases was similar to that with more detailed likelihood data and the estimation of recruitment and stock biomass was also similar.

While there are indications that these implementations are too stiff with respect to growth, disaggregating the commercial fleet and/or allowing for variations in growth rate (both of which are currently possible within the Gadget framework) may help resolve some of these issues.

The potential for the initial parameter values to affect the weighting of survey likelihood components (in the model formulation used here) and in turn the final parameter estimates needs to be considered further. In this case, both the pattern and level of recruitment was the same but given very conflicting information from different surveys this might not be the case.

As an extension to this work, there is a need to demonstrate the methodology for the more complex multi-area case.

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